INDEPENDENT | END-TO-END ANALYTICS

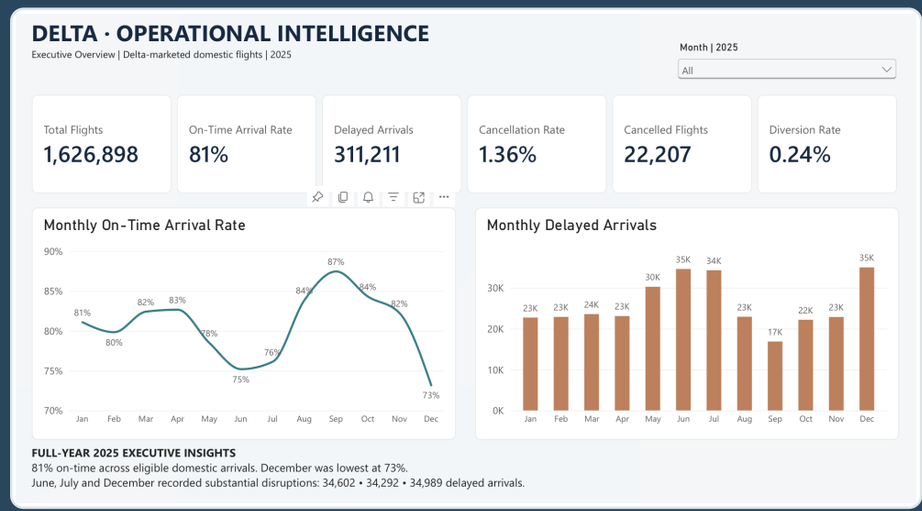
DELTA AIR LINES

Operational & Financial Intelligence

From 1.63 million domestic flight records to executive reporting, operational insights and FP&A decisions.

2025 DOMESTIC OPERATIONS

2021-25 FINANCIALS



1.63M

Delta-marketed domestic flights

80.56%

On-time eligible arrivals

$63.36B

FY2025 consolidated revenue

AAMIR SOHAIB / DATA ANALYTICS + FINANCIAL PERFORMANCE

GITHUB PROJECT →

DELTA · OPERATIONAL INTELLIGENCE

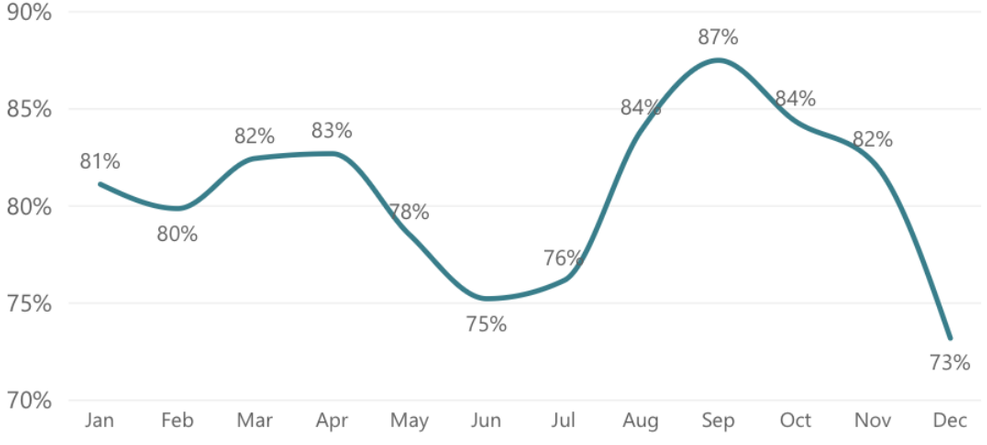
Executive Overview | Delta-marketed domestic flights | 2025

Month | 2025

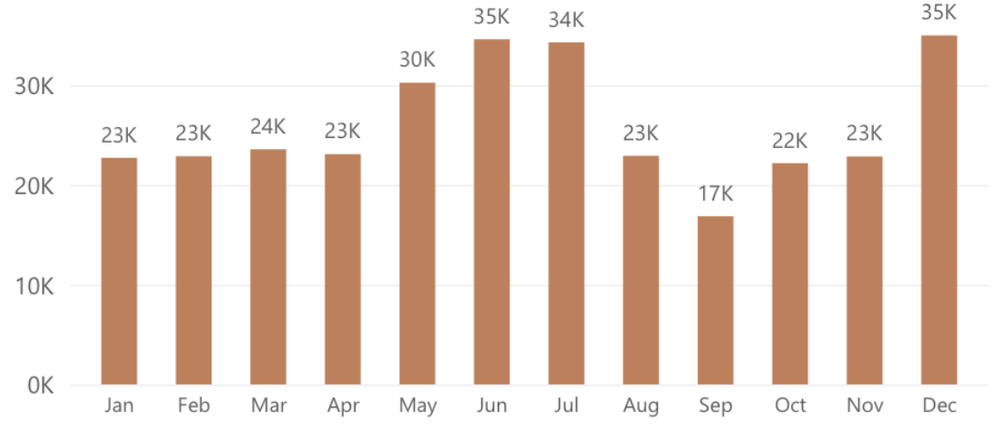
All

Total Flights	On-Time Arrival Rate	Delayed Arrivals	Cancellation Rate	Cancelled Flights	Diversion Rate
1,626,898	81%	311,211	1.36%	22,207	0.24%

Monthly On-Time Arrival Rate



Monthly Delayed Arrivals



FULL-YEAR 2025 EXECUTIVE INSIGHTS

81% on-time across eligible domestic arrivals. December was lowest at 73%.

June, July and December recorded substantial disruptions: 34,602 • 34,292 • 34,989 delayed arrivals.

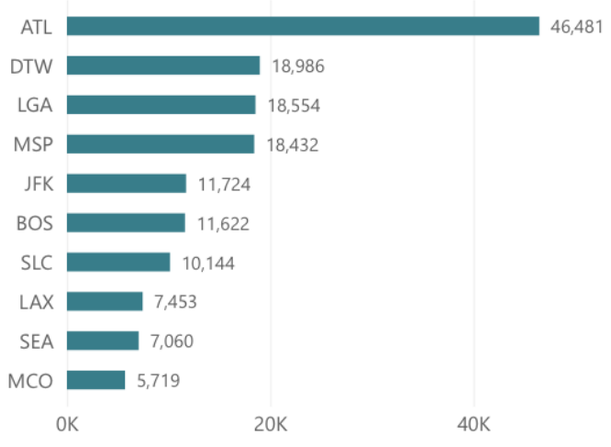
DELTA • OPERATIONAL INTELLIGENCE

Airport, Route & Delay Intelligence | Delta-marketed domestic flights | 2025

Month | 2025

All

Top 10 Destination Airports by Delayed Arrivals



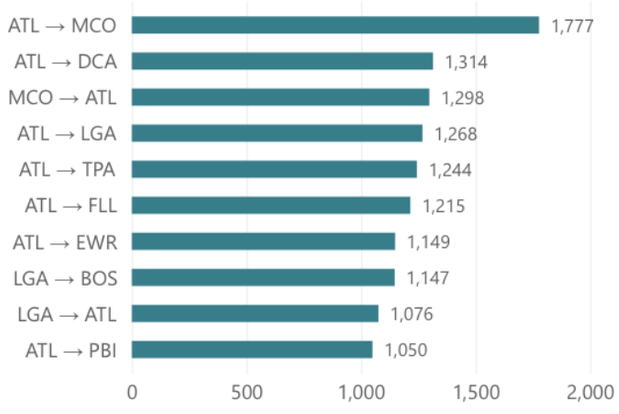
Airports with Elevated Arrival Delay Rates

≥1,000 eligible arrivals in selected period

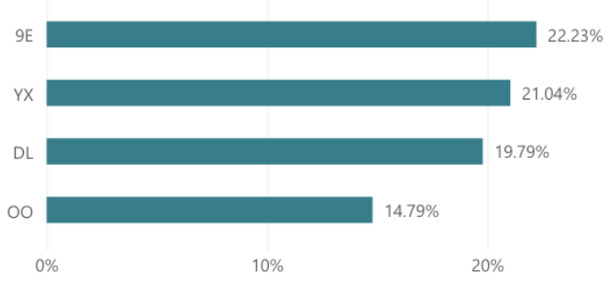
Airport	Delay rate	Eligible arrivals
EWR	34.06%	8,655
PBI	28.29%	7,070
DCA	28.01%	19,078
ORD	27.84%	13,474
DFW	27.36%	11,843
MIA	27.20%	11,656
DEN	26.82%	11,925
MCO	26.63%	21,475
FLL	26.32%	12,365

Top 10 Routes by Delayed Arrivals

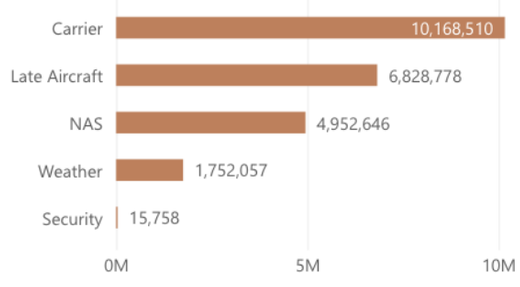
Top 10 in selected months · annual leaders ≥500 eligible



Arrival Delay Rate by Operating Carrier



Reported Delay Minutes by Category



FULL-YEAR 2025 FINDINGS

ATL accounted for 14.94% of network delayed arrivals.

LGA combined substantial delay volume with an above-average arrival delay rate.

Carrier and Late Aircraft represented 71.66% of reported delay minutes.

Airport and directional route patterns vary by location and season.

Carrier codes: DL Delta · OO SkyWest · 9E Endeavor · YX Republic. Comparisons reflect different routes, airports and schedules.
Reported delay minutes are attributed categories; they are not unique disrupted flights or financial impacts.

DELTA • FINANCIAL INTELLIGENCE

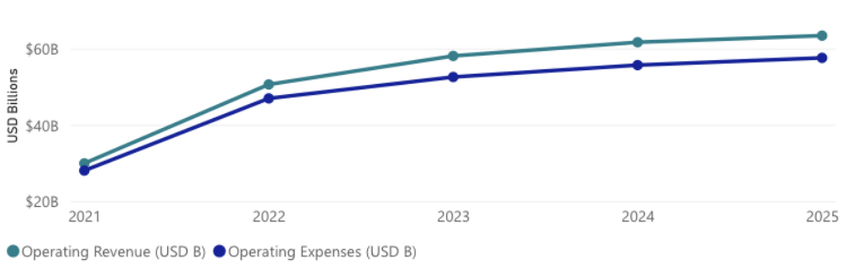
Consolidated Financial Performance | 2021–2025 | Financial Values in USD Billions

fiscal_year: 2025

Operating Revenue (USD B)	Operating Income (USD B)	Operating Margin %	Operating Expenses (USD B)	Fuel Expenses (USD B)
$63.36B	$5.82B	9.19%	$57.54B	$9.82B

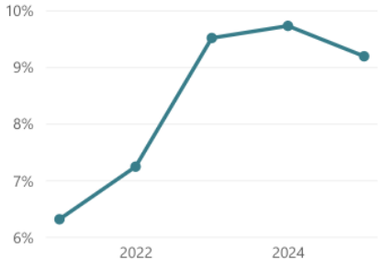
Operating Revenue vs Operating Expenses

2021–2025 • USD billions



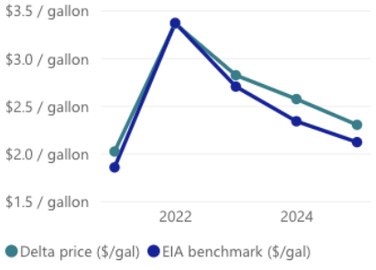
Operating Margin Trend

2025 margin –0.54 pp vs 2024



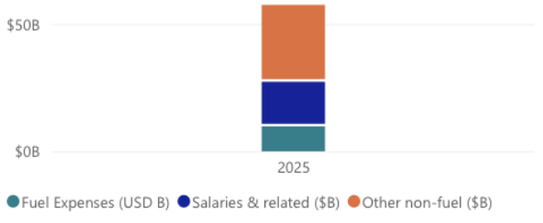
Delta Fuel Price vs EIA Benchmark

USD/gal • External EIA benchmark



Operating Expense Composition

Selected fiscal year • 2025 default • USD billions



WHAT CHANGED IN OPERATING COSTS? | 2024–2025

Fuel expenses	–$0.75B
Salaries and related costs	+$1.36B
Other non-fuel expenses	+$1.28B
Total operating expense rise	+$1.89B

Non-fuel growth more than offset fuel savings.

FINANCIAL EXECUTIVE FINDINGS | FY2025

01 Revenue +2.79%; operating expenses +3.40%.

02 Operating margin: 9.73% → 9.19%.

03 Fuel expenses –$0.75B; non-fuel expenses +$2.64B.

04 Salaries and related costs added $1.36B.

2025 VERIFIED KPI VALUES | Revenue $63.36B • Income $5.82B • Margin 9.19% • Operating expenses $57.54B • Fuel expenses $9.82B
FUEL CONSUMPTION | 2024: 4.11B gal → 2025: 4.27B gal (+3.77%); reported fuel expense fell 7.07%.
Delta Air Lines consolidated results, not allocated to domestic flights. EIA is an external market benchmark, not Delta's paid price.

FY2025

$5.82B
Operating income

9.19%
Operating margin

$57.54B
Operating expenses

Fuel savings did not offset non-fuel cost increases.

From raw data to trusted KPIs

Reproducible data preparation, relational modeling and reporting controls

THREE COMPLEMENTARY DATA SOURCES

01 / OPERATIONS

BTS | 2025

1,626,898 flights

Airport, route, carrier, cancellations and reported delay causes.

02 / FINANCIALS

DELTA / SEC | 2021-25

5 fiscal years

Revenue, operating expense, income and cost composition.

03 / FUEL BENCHMARK

EIA | 2021-25

60 monthly observations

U.S. Gulf Coast spot prices for market context.

END-TO-END IMPLEMENTATION

01
PREPARE
Python / Pandas

›

02
STORE
PostgreSQL

›

03
ANALYZE
SQL + QA

›

04
SERVE
7 reporting views

›

05
REPORT
Power BI + DAX

RECONCILIATION CHECKS

1,600,753 eligible arrivals / 311,211 delayed arrivals / 22,207 cancellations / 3,937 diversions. Report KPIs reconciled to SQL baselines.

Insights that support decisions

Turn descriptive findings into focused operational and financial questions

OPERATIONS

Prioritize network disruption reviews

ATL accounted for 46,481 delayed arrivals, or 14.94% of network delay volume. Pair counts with normalized rates before targeting airports or routes.

FP&A

Explain expense and margin movement

Fuel expense declined $0.747B while salaries and other non-fuel costs rose $2.641B. FY2025 operating margin was 9.19%.

EXECUTIVE REPORTING

Track a consistent performance narrative

On-time arrival performance varied from 87.46% in September to 73.16% in December. Reporting distinguishes delays, cancellations and diversions.

INTERPRETATION BOUNDARIES

Operational scope: 2025 Delta-marketed U.S. domestic flights. Financial scope: Delta consolidated results for FY2021-FY2025. These are distinct analytical populations; no route-level profitability is implied.

EIA jet-fuel spot prices are external benchmarks, not realized purchase prices. Delay categories reflect reported attribution, not proven root cause. Predictive modeling and intervention testing remain future extensions.

Explore the work. Understand the limits.

Public repository, deliverables and the reason the editable PBIX is withheld

PUBLIC REPOSITORY

github.com/roceye1/Delta-Operational-Intelligence

Open source code, reporting data, dashboard images, PDF and methodology

WHAT THE REPOSITORY INCLUDES

Python + SQL
Cleaning, validation, database schema and analysis scripts

Data documentation
Source register, methodology, dictionary and QA evidence

Dashboard evidence
Three images + seven reporting exports and an Excel workbook

REPORTING ASSETS & NOTES

Power BI showcase
Static visuals, not a shareable editable report

DAX transparency
52 base measures + 7 display measures reported; source not fully exported

Planned extensions
Predictive notebook development is not yet complete

WHY THE ORIGINAL .PBIX FILE IS NOT PUBLIC

The original Power BI file contains an account-specific university SharePoint/OneDrive connection URL. It is excluded to protect identifying connection metadata. This public showcase provides static dashboard exports and supporting code/data instead; it does not reproduce interactive Power BI editing or slicers.